



BIOCOMPUTATIONAL ENGINEERING

ENBC321/BIOE658I: Machine Learning for Data Analysis Spring 2026

Credits: 3

Prerequisites: ENBC332, ENBC311

Instructor

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Biocomputational Engineering

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Meeting times and location

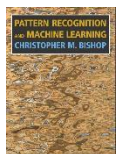
Lecture: BSE-IV Classroom 3335, Mondays 5:00-7:30 PM

Office hours: BSE-IV 4115, Mon/Wed 12:00-1:00 PM (or email to schedule an appointment)

Course Description

This course will teach students the fundamentals of machine learning methods through examples drawn from biological phenomena and clinical data analysis. It is designed to share knowledge of real-world data science and to simplify the learning of complex machine learning theory, algorithms, and coding libraries. The structure of this course provides a step-by-step introduction to the world of machine learning. Major topics covered in this course include **supervised learning** (e.g., regression, classification), **unsupervised learning**, **deep learning**, **dimensionality reduction**, **model selection**, and **model evaluation**. The course is packed with practical machine learning exercises based on real-life examples. Students will not only learn the theoretical aspects of machine learning but also gain hands-on experience building models using programming tools such as Python and R.

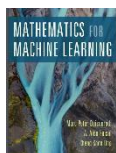
Required Textbooks



Pattern recognition and machine learning

Christopher M. Bishop

Springer, 2006.



Mathematics for machine learning

Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong

Cambridge University Press, 2020.

Course Learning Outcomes

In this course, you will

- Deeply understand the concepts of supervised and unsupervised learning models and their differences in data processing and decision-making.
- Implement machine learning algorithms using Python or R.
- Apply advanced techniques such as dimensionality reduction to analyze high-dimensional data.
- Select appropriate machine learning models for a wide range of problems or enhance their efficiency through reinforcement learning concepts.
- Efficiently manage data to evaluate various machine learning models across diverse input datasets.

Class Policies

Lectures

- Lecture notes will be posted on **ELMS-Canvas** prior to each class. These notes will serve as the basis for exam material and homework (HW). You are highly encouraged to visit the Canvas site regularly to stay up-to-date with the posted materials.

Communication: Canvas or office hours ONLY

- I encourage you to speak with me during office hours or via ELMS email. If a student is not able to attend office hours, meetings by appointment can be scheduled.
- I will answer ELMS communication as soon as possible, definitely within 24 hours on a weekday or within 48 hours on the weekend. ELMS is an essential part of the course, and you should master the use of this resource.
- All questions related to the course whether content, homework, or etc. (except personal) must be posted on **ELMS-Canvas**, so all students can read the question and answers. I will answer the questions as fast as possible. Students are encouraged to post answers to these questions, and I will comment afterwards.
- Please send me an email in case you have administrative questions or something urgent/important related to the course.

Attendance

- Attendance is highly recommended as material covered in class will be the backbone of homework and exams.
- It is also highly recommended that students review the assigned textbook material before class.

Assignments

- Homework assignments (30%) are designed to guide students in conducting a successful research project, providing practical experience in applying research methods and techniques.
- Homework assignments will be posted to ELMS and the assignments will be due at the beginning of class on the date listed on the homework.
- Late homework can be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Quizzes

- Quizzes (10%) will emphasize the topic given in the corresponding week.
- Expect the quizzes every **Wednesday**.
- There will be no make-up for missed quizzes. You will be given one of the quizzes as a “freebie” in case you are sick, etc.

Course evaluation

- Student feedback on lectures every **Wednesday after the quiz**.

Exams

- The midterm exam will cover all content taught before the midterm.
- The project will be assigned before the midterm.
- Project presentations will take place during the last class session.
- The final exam will be cumulative, covering material from the entire course.

Grading

30% - Assignments

10% - Quizzes

20% - Midterm

20% - Group Project

20% - Final Exam

Grading Scale

The instructor will use a “+/-” grading scheme for final grades.

Grade	% Attained
A-, A, A+	90 – 100
B-, B, B+	80 – 89
C-, C, C+	70 – 79
D-, D, D+	60 – 69
F	< 60

Important Note: This syllabus, along with course assignments and due dates, are subject to change. It is the student’s responsibility to check ELMS for corrections or updates to the syllabus.

Tips on How to Do Well

- **Come to lecture, pay attention/take notes, and do the assigned readings** *A lot of content will be covered in this class and it will be very hard to cram the material the night before an exam. The easiest way to make sure you are staying up-to-date with the material is to simply come to class and do the reading during the week of the lecture.*
- **Do the assignments.** *The assignments will give you an idea of what the exam will look like. Do them on time.*
- **Form a study group.** *Many new concepts will be covered before each exam. Form a study group to discuss the material and test one another. Work together to make sure you have a comprehensive understanding of the material.*

Accessibility and Disability Services: The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, on the basis of disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. [Accessibility & Disability Service \(ADS\)](#) provides reasonable accommodations to qualified individuals to provide equal access to services, programs and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to make arrangements.

For assistance in obtaining accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at adsfrontdesk@umd.edu. Information about [sharing your accommodations with instructors](#), [note taking assistance](#) and more is available from the [Counseling Center](#).

USG Disability Support Services: The USG Disability Support Services offices acts as a liaison between students at USG and their home campuses to ensure that they have access to their approved accommodations. The USG DSS offices collaborate with faculty, students, and staff members to implement accommodations at USG in a way that preserves confidentiality, equality, and integrity for all students with disabilities to ensure an accessible educational experience for students at USG. The USG DSS offices provide various accommodations to students with disabilities who are registered through their home campus DSS offices. Students must first be registered through their home DSS offices in order to receive accommodations at USG. Once students are registered, specific accommodations are based on documentation of the disability and personal interviews with their home campus DSS coordinators.

For further assistance regarding the USG DSS process, contact USG DSS at usg-disability@umd.edu or call 301-738-6384. Please also visit the USG DSS website at <https://shadygrove.umd.edu/student-services/center-for-academic-success/dss>

Student Resources and Services: Students are encouraged to use academic and student service centers (<https://shadygrove.umd.edu/student-services>) available on the Universities at Shady Grove campus:

- Career & Internship Services Center – provides a wide variety of services to support career readiness, including career coaching and experiential learning opportunities.

- Center for Counseling and Consultation - employs licensed therapists who provide free and confidential services to enhance the well-being of students, faculty, and staff.
- Center for Academic Success - offers academic coaching, writing consultations, peer-led programs, disability support, and workshops to help students achieve their academic goals.
- Center for Student Engagement and Financial Resources - offers services and programs that promote student success including: student life and organizations, scholarship and financial aid support, USG IDs, and leadership development.

Honor Code: The University of Maryland has a nationally recognized Code of Academic Integrity, administered by the Student Honor Council. This Code sets standards for academic integrity in Maryland for all undergraduate and graduate students. As a student, you are responsible for upholding these standards for this course. It is very important for you to be aware of the consequences of cheating, fabrication, facilitation, and plagiarism. For more information on the Code of Academic Integrity or the Student Honor Council, please visit <http://www.studenthonorcouncil.umd.edu/whatis.html>.

In particular, the Code of Academic Integrity prohibits students from cheating on exams, plagiarizing papers, submitting the same paper for credit in two courses without authorization, buying papers, submitting fraudulent documents, and forging signatures. The University also has adopted an Honor Pledge that should be written and signed on all submitted papers. The Pledge reads: I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination.

Policy on Religious Holidays: The University System of Maryland policy on religious observances provides that “students should not be penalized because of observances of their religious beliefs; students shall be given an opportunity, whenever feasible, to make within a reasonable time any academic assignment that is missed due to individual participation in religious observances.” However, “it is the student's responsibility to inform the instructor of any intended absences for religious observances in advance. Notice should be provided as soon as possible but no later than the end of the schedule adjustment period.”

UMD Course Related Policies: For further information about the University of Maryland policies for undergraduate students, please visit <http://www.ugst.umd.edu/courserelatedpolicies.html>

Schedule (subject to change)

Week	Date	Topic
1	January 26	History of AI; Introduction to Machine Learning; Mathematical Foundations for Machine Learning: Linear Algebra
2	February 2	Mathematical Foundations for Machine Learning: Linear Algebra; Statistics; Data Summary, Probability, Normal Distribution, Binomial Distribution, Describing Relationships; Data Processing: Data Cleaning; Train Test Split
3	February 9	Supervised Learning: Regression; Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Feature Selection Methods Advanced Regression: Support Vector Regression (SVR)
4	February 16	Advanced Regression: Support Vector Regression (SVR); Decision Tree Regression; Random Forest Regression; Entropy
5	February 23	Classification: Logistic Regression; K-Nearest Neighbors (K-NN); Linear Support Vector Machine (SVM)
6	March 2	Review for the MIDTERM Exam
7	March 9	Classification: Non-Linear Support Vector Machine (SVM) MIDTERM – Everything covered so far
8	March 16	Spring Break (March 15-22)
9	March 23	Classification: Bayes' Theorem, Naïve Bayes Theorem; Decision Tree Classification; Random Forest Classification Guest Speaker: TBD
10	March 30	Model Selection; Model Evaluation; Cumulative Accuracy Profile (CAP); Receiver Operating Characteristics (ROC)
11	April 6	Unsupervised Learning: Clustering: K-Means Clustering, Hierarchical Clustering
12	April 13	Dimensionality Reduction: Principal Component Analysis (PCA)
13	April 20	Deep Learning: Introduction to Neural Networks; Artificial Neural Networks (ANN); The Neuron; The Activation Functions; How do ANNs work?; How do ANNs learn?; Gradient Descent; Stochastic Gradient Descent
14	April 27	Deep Learning: Convolutional Neural Networks (CNN); Convolutional Layers; Pooling Layers; Flatten Layers; Full Connection
15	May 4	Review for the FINAL Exam Project Presentations
	May 11-18	FINAL EXAM – All course materials